

PSP Downtime Database Activity Runbook

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Goal

- Setup Oracle Hash to Oracle Range (Full Load + CDC Replication) with minimal downtime and minimal FCI. Activity details are shared [here](#).
- Add Aurora Postgres global for Audit database through IPS and also resolve drift changes (QPM, Instance roles)
- Upgrade IPS CDK version from 0.6.14 to 0.8.4 (Optional based on outcomes from preprod)

Summary

Status

Environment	Production
Status	COMPLETED
Slack Channel	#psp-prod-db-hash-to-range-replication
JIRA	PSP-26356
Activity Details	Setup GoldenGate Replication - Oracle Hash --> Oracle Range

Impacted Assets

Application	Dev Portal Link
PSP	PSP Dev Portal
PAPI	PAPI Dev Portal
PubSub	Pubsub Dev Portal
RJF	RJF Dev Portal
PayrollIDM	PayrollIDM Dev Portal

CR Details

Application	Change Request	Images	Status
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PSP	CHG2229163	master-ce820f2-eks	CLOSED
PAPI	CHG2229170	master-50894fe	CLOSED
Pubsub	CHG2229203	master-0e656f6	CLOSED
RJF	CHG2229215	master-3e1dda8	CLOSED
Payroll DM	CHG2229206	master-eebc5a6	CLOSED

Maintenance Windows

S.No	Application	A2I Maintenance URL
1	DT-Payroll-Svc	https://intuit.wavefront.com/maintenancewindows/1684585778332
2	QBDT-Payroll-Service	https://intuit.wavefront.com/maintenancewindows/1684585951522
3	CEP	https://intuit.wavefront.com/maintenancewindows/1684586213557
4	Case 360	https://intuit.wavefront.com/maintenancewindows/1684586392548
5	EWS	https://intuit.wavefront.com/maintenancewindows/1684586541304
6	PTC	https://intuit.wavefront.com/maintenancewindows/1684587194468
7	QBDT Payroll Entitlement	https://intuit.wavefront.com/maintenancewindows/1684587307963
8	Workforce	https://intuit.wavefront.com/maintenancewindows/1684587468515
9	BRM	https://intuit.wavefront.com/maintenancewindows/1684587593115
10	Desktop Pay API	https://intuit.wavefront.com/maintenancewindows/1684587763643
11	Payroll DM	https://intuit.wavefront.com/maintenancewindows/1684587894051
12	Pubsub	https://intuit.wavefront.com/maintenancewindows/1684588413633

Duration

Step	Expected Downtime	Activity Start Time	Activity End Time	Duration
Bring down application	~15 mins	21 May 2023 11AM IST		

Downtime Details

Application	Expected Downtime	Actual Downtime	Downtime Start (IST)	Downtime End (IST)
PSP	100 Mins	72 Mins	11: 31 AM	12:43 PM
PAPI	120 Mins	87 Mins	11: 29 AM	12: 56 PM
Pubsub	120 Mins	87 Mins	11: 29 AM	12: 56 PM
Payroll DM	120 Mins	87 Mins	11: 29 AM	12: 56 PM
Monolith DB (Backup + IPS)	60 mins	40 Mins	11: 39 AM	12:20 PM

Important Resources

JSS

Application	JSS Link
PSP Batch Jobs	PSP JSS View

RJF Reports	RJF JSS View
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Splunk

Type	Link
Application	Splunk Application Production
Gateway	Splunk Gateway Production

ArgoCd

Environment	Application	ArgoCD Link
Production	PSP	Production PSP ArgoCd
	PAPI	Production PAPI ArgoCd
	Pubsub	Production ArgoCd
	RJF	Production RJF ArgoCd
	PayrollDM	Production PayrollDM ArgoCd

Notification Email Addresses

 SBSEG-PD QBDT PSP AWS <SBSEG-PDQBDTPSPAWS@intuit.com>; SBG PSP Services Decom Scrum <SBGPSPServicesDecompScrum@intuit.com>; SBSEG-PD-Iqbal-Directs <SBSEG-PDIqbalDirect@intuit.com>; SBSEG-PD-Kamesh-Directs <SBSEG-PD-Kamesh-Directs@intuit.com>; SBSEG SBG Payroll PSP Informed <SBGPayrollPSPInformed@intuit.com>; SBSEG-Iqbal-Extended <SBSEG-PDIqbalExtended@intuit.com>; SBSEG-PD DTPayrollExtended <SBSEG-PDDTPayrollExtended@intuit.com>; SBSEG-PD DTPayroll <SBSEG-PDDTPayroll@intuit.com>

Runbook

Pre-Activity Steps

S. No	Task	Steps	Status	Owner
1	Inform Parter Teams	Refer NotificationEmailAddresses Section.	COMPLETED	N, Ramasub ramani
2	Pause Ledger Balance Job	Pause Ledger Balance Job from PSP JSS View	COMPLETED	Sur, Koustav
3	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. #payroll-pd-releases #sbg-payroll-prod #ioc #psp-rtb-priority	COMPLETED	N, Ramasub ramani
4	Reassign PagerDuty.	Update PagerDuty Schedule. Pager Duty Schedules : SBG - QBPY - DPS : DevOps PSP (IDC)	COMPLETED	N, Ramasub ramani
5	Note down all the PSP/RJF JSS Jobs in the same window.	Link : PSP JSS View Batch Jobs : Batch jobs RJF Jobs : RJF	COMPLETED	Sur, Koustav

6	Run Sanity at 11AM for PSP /PAPI/Pubsub/RJF.	Execute below Sanity Jobs to ensure jobs working fine as expected. Refer PSP AWS TipToe - PROD Sanity Tests to generate tokens. <table><tr><th>App</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td> PASSED </td></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td> PASSED </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> PASSED </td></tr><tr><td>RJF</td><td>RJF Sanity</td><td> PASSED </td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td> PASSED </td></tr></table>	App	Sanity Link	Status	PSP	PSP Sanity	PASSED	PAPI	PAPI Sanity	PASSED	Pubsub	Pubsub Sanity	PASSED	RJF	RJF Sanity	PASSED	PayrollDM	PayrollDM Sanity	PASSED	COMPLETED	Hoolimath, Praveenkumar
App	Sanity Link	Status																				
PSP	PSP Sanity	PASSED																				
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Pubsub	Pubsub Sanity	PASSED																				
RJF	RJF Sanity	PASSED																				
PayrollDM	PayrollDM Sanity	PASSED																				
7	Bring Down Staging environment	Use ZeroPodDeploymentPR for Middleware	COMPLETED	Hoolimath, Praveenkumar																		
8	Ensure all Change Requests are scheduled.	Refer CRDetails section. Navigate to each CR and ensure CRs are scheduled.	COMPLETED	N, Ramasubramani																		
9	Bootstrap us-east-2 region via IPS	generate temp credentials cdk bootstrap aws://893547637742/us-east-2 -c env=prod -c group=dr-ibob -c config=psp-prerequisites-prod-ue.json	COMPLETED	Narlagalla, Praveen Goud																		

Activity Steps

S. No	Task	Steps	Status	Owner	Remarks														
1	Ensure there are no critical jobs running.	- Navigate to JSS View Link. - Wait for critical jobs to complete.	COMPLETED	Sur, Koustav															
2	Ensure A2I Maintenance alerts are snoozed	Refer MaintenanceWindows Section.	COMPLETED	Sur, Koustav															
3	Bring down below apps. 1. PayrollDM 2. Pubsub 3. RJF 4. PAPI	- Use ZeroPodDeploymentPR for applications except PSP. - Wait for POD count to become zero.	COMPLETED	Hoolimath, Praveenkumar															
4	Bring down PSP along with Maintenance Page	- Use ZeroPodDeploymentPR for PSP. - Wait for POD count to become zero.	COMPLETED	Hoolimath, Praveenkumar															
5	Ensure all apps are down.	Hit following URLs for each application. Refer PSP AWS TipToe - PROD Sanity Tests to generate tokens. <table><tr><th>App</th><th>Sanity</th></tr><tr><td>PSP</td><td>PSP Sanity</td></tr><tr><td>SAP</td><td>SAP</td></tr><tr><td>PAPI</td><td>PAPI Sanity</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td></tr><tr><td>RJF</td><td>RJF Health Check</td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td></tr></table>	App	Sanity	PSP	PSP Sanity	SAP	SAP	PAPI	PAPI Sanity	Pubsub	Pubsub Sanity	RJF	RJF Health Check	PayrollDM	PayrollDM Health Check	COMPLETED	Sur, Koustav	
App	Sanity																		
PSP	PSP Sanity																		
SAP	SAP																		
PAPI	PAPI Sanity																		
Pubsub	Pubsub Sanity																		
RJF	RJF Health Check																		
PayrollDM	PayrollDM Health Check																		
6	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	COMPLETED	Sur, Koustav															
7	Note down All the failed RJF /PSP BatchJobs.	Batch Jobs : RJF Jobs :	COMPLETED	Sur, Koustav															

8	Lock Application Users	alter user PSPADM account lock; alter user PSPAPP account lock;	COMPLETED	Choubey, Mayank	
9	Kill sessions (if needed)	- Wait for Inflight requests/Running Batch Jobs to complete - time boxed to 2 mins. - There should not be any ACTIVE connections to Database. Monolith <pre>select 'exec rdsadmin.rdsadmin_util.kill ' s.sid ',' s.serial# ',' ''PROCESS'' ');' from gv\$session s where USERNAME='PSPAPP'; select 'exec rdsadmin.rdsadmin_util.kill ' s.sid ',' s.serial# ',' ''PROCESS'' ');' from gv\$session s where USERNAME='PSPADM';</pre> Audit: <pre>select pg_terminate_backend(pid) from pg_stat_activity where pid in (select pid FROM pg_stat_activity where username in ('ibob_prod_pspapp','ibobadm_owner','ibobadm_readonly '));</pre>	COMPLETED	Choubey, Mayank	
10	Ensure there are no active connections to the Database.	No active connections from PSP / PAPI / Pubsub / RJF / PayrollDM. Monolith: <pre>set lines 300 col username for a30 col machine form a70; select username,machine,count(*) from gv\$session where username is not NULL and username not in ('RDSADMIN','INTUADMIN','SYS','SYSTEM') and type = 'USER' group by service_name,username, machine order by username,machine;</pre> Audit: <pre>SELECT datname as db,username as username,client_addr as machine, count(*) FROM pg_stat_activity where username not in ('rdsadmin','postgres') group by db,username,machine;</pre>	COMPLETED	Choubey, Mayank	
11	Check GG Replication Lag	Wait till Replication lag to zero.	COMPLETED	Choubey, Mayank	
12	Create Extract and Pump on Hash GG hub	<pre>obey ./diroby/create_epsphp02.oby obey ./diroby/dblogin_s_psphpp02.oby register extract epsphp02 obey ./diroby/create_ppsprp01.oby</pre>	COMPLETED	Choubey, Mayank	
13	Take Database Snapshot	Take RDS Snapshot via aws rds console. Wait for it to be completed.	COMPLETED	Choubey, Mayank	

14	Configure Aurora Postgres Global database for Audit Postgres	IPS Steps <pre> eiamCli aws_creds -a 893547637742 -r PowerUser -p default #Change region in credentials file accordingly #Bootstrap for region (if required) cdk bootstrap aws://893547637742/us-east-2 -c env=prod -c group=dr-ibob -c config=config/psp- prerequisites-prod-ue.json #Prod US-west #list cdk list --all -c env=prod -c group=primary-ibob -c config=config/psp-prerequisites-prod.json cdk list --all -c env=prod -c group=primary-ibob -c config=config/psp_ibob-prod.json #diff cdk diff --all -c env=prod -c group=primary-ibob -c config=config/psp-prerequisites-prod.json cdk diff --all -c env=prod -c group=primary-ibob -c config=config/psp_ibob-prod.json #Deploy cdk deploy --all -c env=prod -c group=primary-ibob - c config=config/psp-prerequisites-prod.json cdk deploy --all -c env=prod -c group=primary-ibob - c config=config/psp_ibob-prod.json #DR Us-east #list cdk list --all -c env=prod -c group=dr-ibob -c config=config/psp-prerequisites-prod-ue.json cdk list --all -c env=prod -c group=dr-ibob -c config=config/psp_ibob-prod.json #diff cdk diff --all -c env=prod -c group=dr-ibob -c config=config/psp-prerequisites-prod-ue.json cdk diff --all -c env=prod -c group=dr-ibob -c config=config/psp_ibob-prod.json #deploy cdk deploy --all -c env=prod -c group=dr-ibob -c config=config/psp-prerequisites-prod-ue.json cdk deploy --all -c env=prod -c group=dr-ibob -c config=config/psp_ibob-prod.json </pre>	COMPLETED	Narlagalla, Praveen Goud	
15	Validation steps for IPS aurora global and drift changes	• Verify Global DB is up and able to connect • Verify Drift changes from aws console • Verify CFT creation is successful	COMPLETED	Narlagalla, Praveen Goud	
16	Ensure Extract trails has no changes	stats epsphp02	COMPLETED	Choubey, Mayank	
17	Unlock application users	<pre> alter user PSPAPP account unlock; alter user PSPADM account unlock; </pre>	COMPLETED	Choubey, Mayank	

18	Validate application connectivity via application users	Monolith pspapp/"xxxxx"@'pspuwp01.sbg-ppsp-prod.a.intuit.com:1521/pspuwp01' pspadm/"xxxxx"@'pspuwp01.sbg-ppsp-prod.a.intuit.com:1521/pspuwp01' Audit (connect from local/laptop) psql -h localhost -d ibob_prod_pspapp -U prodapgib -p 10539	COMPLETED	Choubey, Mayank										
19	Monitor DB Connections.	Ensure connections are originating from the enabled PSP POD. Monolith set lines 300 col username for a30 col machine form a70; select username,machine,count(*) from gv\$session where username is not NULL and username not in ('RDSADMIN', 'INTUADMIN', 'SYS', 'SYSTEM') and type = 'USER' group by service_name,username,machine order by username,machine; Audit SELECT datname as db,username as username,client_addr as machine, count(*) FROM pg_stat_activity where username not in ('rdsadmin','postgres') group by db,username,machine;	COMPLETED	Choubey, Mayank										
20	Scale up PSP Pods with Maintenance.	- Use ScaleFullDeploymentPR for PSP with Maintenance On. - Sync may21-scale-full-maintenance-on which will bring up all the Pods with Maintenance.	COMPLETED	Hoolimath, Praveenkumar										
21	Monitor Nodes to ensure correct image is deployed	- Monitor Node Level Sanity Dashboard to ensure all nodes are up. - Validate the image imageDetails.sh (Command : ./imageDetails.sh prd)	COMPLETED	Sur, Koustav										
22	Scale up PSP Pods without Maintenance.	- Use ScaleFullDeploymentPR for PSP with Maintenance Off. - Once all Pods are up, Merge and Sync may21-scale-full-maintenance-off - Ensure Maintenance is removed. Refer ArgoCd section. - Make sure there are no pod restarts.	COMPLETED	Hoolimath, Praveenkumar										
23	Run the PSP Sanity.	Execute below Sanity Jobs to ensure apps working fine as expected. <table><tr><th>App</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td></td></tr><tr><td>SAP</td><td>SAP Sanity</td><td></td></tr></table>	App	Sanity Link	Status	PSP	PSP Sanity		SAP	SAP Sanity		COMPLETED	Sur, Koustav	
App	Sanity Link	Status												
PSP	PSP Sanity													
SAP	SAP Sanity													
24	Monitor Overall database /replication health vi CW dashboard	DB Health: DB CW Dashboard GG Lag: GG CW Dashboard	COMPLETED	Choubey, Mayank										
25	Bring up all pods for below services 1. PAPI 2. RJF 3. Pubsub 4. PayrollDM	- Merge ScaleFullDeploymentPR for PAPI/RJF/Pubsub/PayrollDM. - Ensure correct image is deployed. Refer ArgoCd section.	COMPLETED	Hoolimath, Praveenkumar										

26	Run PAPI/Pubsub/RJF/PayrollDM Sanity.	Execute below Sanity Suites. Refer PSP AWS TipToe - PROD Sanity Tests to generate tokens. <table><thead><tr><th>App</th><th>Sanity Link</th><th>Status</th></tr></thead><tbody><tr><td>PAPI</td><td>PAPI Sanity</td><td></td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td></td></tr><tr><td>RJF</td><td>RJF Sanity</td><td></td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td></td></tr></tbody></table>	App	Sanity Link	Status	PAPI	PAPI Sanity		Pubsub	Pubsub Sanity		RJF	RJF Sanity		PayrollDM	PayrollDM Sanity		COMPLETED	Sur, Koustav	
App	Sanity Link	Status																		
PAPI	PAPI Sanity																			
Pubsub	Pubsub Sanity																			
RJF	RJF Sanity																			
PayrollDM	PayrollDM Sanity																			
27	Ensure we see DB Connections from PSP/Pubsub/PAPI/RJF/PayrollDM.	Monolith <pre>set lines 300 col username for a30 col machine form a70; select username,machine,count(*) from gv\$session where username is not NULL and username not in ('RDSADMIN','INTUADMIN','SYS','SYSTEM') and type = 'USER' group by service_name,username,machine order by username,machine;</pre> Audit <pre>SELECT datname as db,username as username,client_addr as machine, count(*) FROM pg_stat_activity where username not in ('rdsadmin','postgres') group by db,username,machine;</pre>	COMPLETED	Choubey, Mayank																
28	Resume Ledger Balance Job	Resume Ledger Balance Job from PSP JSS View .	COMPLETED	Sur, Koustav																
29	Run PSP & RJF failed jobs and missed Jobs.	Run PSP & RJF failed jobs and missed Jobs. Run LedgerBalance Job if Missed.	COMPLETED	Sur, Koustav																
30	Validate no jobs are stuck	- Check if any RJF/PSP Batch Jobs running for a long time. - Check if the pod is still active. If pod is inactive, mark the job to completed.	COMPLETED	Sur, Koustav																

Post Activity Steps

	Task	Steps	Status	Owner	Remarks
1	Monitor all the apps and databases.	Refer Monitoring Section.	COMPLETED	N, Ramasubramani Sur, Koustav Hoolimath , Praveenkumar	
2	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	COMPLETED	Sur, Koustav	
3	Resume/rollback A2I snoozed alerts	Refer MaintenanceWindows Section.	COMPLETED	Sur, Koustav	
4	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	COMPLETED	N, Ramasubramani	
5	Close All open Change Requests.	Update CRDetails Section.	COMPLETED	N, Ramasubramani	
6	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	COMPLETED	Hoolimath, Praveenkumar	
7	Inform partner teams slack channel.	Inform following Slack Channels. #payroll-pd-releases #sbg-payroll-prod #ioc #psp-rtb-priority	COMPLETED	N, Ramasubramani	

Deployment PRs

Zero Pod Deployment PR

Application	PR	Image	Operation	Owner	Reviewer	Status
PSP Staging	feature-May21-scale-to-zero		Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
PAPI Staging	may21-scale-to-zero-stg		Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
Parallel Setup MW	may21-scale-to-zero		Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
PayrollDM	may21-scale-to-zero	master-eebc5a6	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
Pubsub	may21-scale-to-zero	master-0e656f6	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
RJF	may21-scale-to-zero	master-3e1dda8	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
PAPI	may21-scale-to-zero	master-50894fe	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
PSP (with maintainence ON)	may21-scale-to-zero	master-ce820f2-eks	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED

One Pod Deployment PR

Application	PR	Operation	Owner	Reviewer	Comments
PSP (with maintainence ON)	may21-scale-to-zero		Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	Will be bringing up all pods at once

Scale Full Deployment PR (Merge and Sync)

Application	PR	Image	Operation	Owner	Reviewer	Status
PSP	may21-scale-full-maintainence-on (with maintainence ON)	master-ce820f2-eks	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
	may21-scale-full-maintainence-off (with maintainence OFF)	master-ce820f2-eks	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
PAPI	environments/prd-usw2-eks	master-50894fe	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
Pubsub	environments/prd-usw2-eks	master-0e656f6	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
RJF	environments/prd-usw2-eks	master-3e1dda8	Sync	Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED
PayrollDM	environments/prd-usw2-eks	master-eebc5a6		Hoolimath, Praveenkumar	N, Ramasubramani / Joshi, Vishwanath	COMPLETED

Learnings

S No	Learning	Action Item	Owner	Status	Comments
1	One click Jenkins job for bringing down/single pod /full scale the entire application		N, Ramasubramani / Muthuram, Karthikeyan	OPEN	
2	Integrate Praveen's imageDetails.sh & ConfigMap into the job & deployment job		N, Ramasubramani / Muthuram, Karthikeyan	OPEN	
3	Triggering a RDS backup a hour before the actual backup reduced the backup time	Add this step to Runbook Pre-Activity steps.	N, Ramasubramani	OPEN	

4	IPS bootstrapping in the us-east-1 region reduced the downtime by 15 mins		Choubey, Mayank	OPEN	
5	Give partner team heads up 1/2 week before	Add this step to Runbook Pre-Activity steps.	N, Ramasubramani	OPEN	
6	IPS CDK apply took more time than expected		Choubey, Mayank	OPEN	
7	Monitoring email was not sent	Add this step to Runbook Post Activity steps.	N, Ramasubramani	OPEN	
8	Sanity and Health Check URL should be different.	• Use Health Check URL when the app is down. • Use Sanity when the apps are up.	Sur, Koustav	OPEN	
9	Payroll DM Health Check URL was not working.		Sur, Koustav	OPEN	